



TITLE: AI-Powered Demand & Inventory Intelligence Platform

Submitted By:

- **Faruqe Ansari**
- **Kairam Konda Sri Bhavya**
- **Erugupally Shiva Kumar**
- **Reejan Kushma**

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ABSTRACT

Inventory mismanagement—whether overstocking or stockouts—remains a significant challenge for businesses across industries. This project, undertaken as an internship at Zidio Development, aims to address this problem through the development of an "AI-Powered Demand & Inventory Intelligence Platform"

The platform utilizes a machine learning-based approach to forecast product demand using historical sales data. Key steps included data collection, preprocessing, feature engineering, model training, and dashboard development. A Random Forest Regressor was selected for its robustness and ability to handle non-linear relationships. The model was evaluated using metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), achieving promising results.

An interactive dashboard was built using Streamlit, providing users with real-time demand forecasts, low-stock alerts, and visual insights into sales patterns. The system enables data-driven decision-making for inventory planning.

The project concludes that AI-powered forecasting can significantly improve inventory accuracy, reduce operational costs, and enhance customer satisfaction. Future enhancements could include integration with live data streams and the addition of external factors such as weather and holidays to improve prediction accuracy.

Keywords: Demand Forecasting, Inventory Management, Machine Learning, Random Forest, Data Visualization, Dashboard, Business Intelligence

DECLARATION

I hereby declare that the project report titled "AI-Powered Demand & Inventory Intelligence Platform"

submitted to Zidio Development as part of my internship is my original work. I confirm that:

1. This project has not been submitted for any other degree, diploma, or internship elsewhere.
2. All the information, data, and results presented in this report are genuine and based on my own efforts.
3. Any external sources, references, or materials used have been properly acknowledged and cited.
4. The project work was carried out under the guidance of my supervisor at Zidio Development.

I take full responsibility for the authenticity and accuracy of the information presented in this report.

Date: August 20, 2026

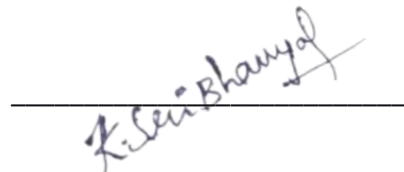
SIGNATURES



Faruq Ansari



Erugupally Shiva Kumar



Kairam Konda Sri Bhavya



Reekan Kushma

CERTIFICATE

This is to certify that

Faruqe Ansari

Kairam Konda Sri Bhavya

Erugupally Shiva Kumar

Reejan Kushma

have successfully completed their internship project titled "AI-Powered Demand & Inventory Intelligence Platform" at Zidio Development during the period of [Start Date] to [End Date] Their work involved the development of a machine learning-based platform for demand forecasting and inventory intelligence. The project was executed with dedication and met the desired objectives.

We appreciate their contribution and wish them success in their future endeavors.

Date: August 20, 2026

Place: [Your City]

SIGNATURE

Name: [Supervisor Name]

Designation: [Supervisor Designation]

Zidio Development

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CHAPTER 1: INTRODUCTION TO THE TOPIC

1.1 Background

Inventory management is a critical function for businesses across all industries. It involves overseeing the flow of goods from manufacturers to warehouses and ultimately to customers. Effective inventory management ensures that a company has the right products, in the right quantities, at the right time, and at the right cost.

However, managing inventory is a complex challenge. Businesses often face two major problems: overstocking (holding too much inventory) and stockouts (running out of stock). Overstocking ties up capital and increases storage costs, while stockouts lead to lost sales and dissatisfied customers.

Traditionally, businesses have relied on manual methods or simple rule-based systems to manage inventory. While these methods are easy to implement, they often fail to capture complex demand patterns, resulting in inefficiencies and missed opportunities.

1.2 The Role Of AI in Demand Forecasting

In recent years, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as powerful tools to address inventory management challenges. Unlike traditional methods, AI models can analyze large volumes of historical sales data, identify hidden patterns, adapt to seasonal change, and provide real-time predictions.

By leveraging AI, businesses can move from reactive inventory management to predictive inventory management, anticipating demand and planning accordingly.

1.3 Problem Statement

At Zidio Development, there was a need for an intelligent system that could accurately forecast product demand, provide early warnings for low-stock items, and enable data-driven decision-making.

ing. Without such a system, the company faced issues like inefficient inventory levels, wasted resources, and missed revenue opportunities.

This project addresses these challenges by developing an AI-powered platform for demand forecasting and inventory intelligence.

1.4 Objectives

The primary objectives of this project are:

1. To design and develop a machine learning model for forecasting product demand based on historical sales data.
2. To build an interactive dashboard that visualizes demand predictions, inventory alerts, and key insights.
3. To provide actionable insights to help businesses optimize inventory levels and reduce costs.
4. To demonstrate the practical application of AI in solving real-world business problems.
5. To deliver a user-friendly platform that can be easily used by non-technical stakeholders.

1.5 Scope

The project includes:

- Data collection from internal sales records
- Development of a Random Forest machine learning model
- Creation of an interactive Streamlit dashboard
- Evaluation using MAE, RMSE, and R^2 metrics

The project does NOT include:

- Live integration with ERP or POS systems
- Financial reporting
- Mobile application development
- Consideration of external factors (weather, holidays, etc.)

1.6 Report Organization

Chapter 2 presents the literature review and technology stack. Chapter 3 describes the system design and methodology. Chapter 4 covers implementation and results. Chapter 5 discusses challenges and future scope. Chapter 6 provides the conclusion.

CHAPTER 2: REVIEW OF LITERATURE & TECHNOLOGY STACK

2.1 Introduction to Inventory Management.

Inventory management is a critical function in supply chain operations. It involves overseeing the flow of goods from manufacturers to warehouses and ultimately to customers. The primary goal is to maintain optimal inventory levels — enough to meet customer demand without holding excess stock.

2.1.1 Introduction to Inventory Management.

Concept	Definition	Purpose
Economic Order Quantity (EOQ)	The optimal order quantity that minimizes total inventory costs	Determine how much to order
Safety Stock	Extra inventory held to prevent stockouts due to demand variability	Buffer against uncertainty
Reorder Point	The stock level at which a new order should be placed	When to reorder
Lead Time	Time taken to receive new stock after placing an order	Plan replenishment
ABC Analysis	Classification of inventory into categories (A, B, C) based on importance	Prioritize management efforts
Inventory Turnover Ratio	Number of times inventory is sold and replaced over a period	Measure efficiency

2.1.2 Traditional Inventory Management Approaches

Approach	Description	Limitations
Manual Tracking	Using spreadsheets or physical counts to track inventory	Time-consuming, error-prone, no real-time visibility
Rule-Based Systems	Using fixed rules (e.g., "order when stock < 100")	Cannot adapt to changing demand patterns
Just-in-Time (JIT)	Ordering stock only when needed	Vulnerable to supply chain disruptions
Economic Order Quantity (EOQ)	Calculating optimal order quantity using a formula	Assumes constant demand, doesn't handle seasonality

2.2 AI & Machine Learning in Demand Forecasting

2.2.1 Why AI for Demand Forecasting?

Traditional forecasting methods often fail to capture complex patterns in sales data. AI and Machine Learning offer several advantages:

Advantage	Description
Handles Non-Linear Relationships	Can capture complex patterns that traditional methods miss
Adapts to Changing Data	Models can be retrained with new data to adapt to changing trends
Handles Large Datasets	Can process and analyze millions of records efficiently

Advantage	Description
Incorporates Multiple Factors	Can consider price, promotions, seasonality, and external factors
Provides Accurate Predictions	Often outperforms traditional statistical methods

2.2.2 Common AI/ML Techniques for Demand Forecasting

Technique	Description	Best Use Case
Linear Regression	Models relationship between dependent and independent variables	Simple, linear relationships
Random Forest	Ensemble of decision trees for robust predictions	Complex, non-linear relationships
XGBoost	Gradient boosting algorithm with regularization	High-performance tabular data
LSTM (Long Short-Term Memory)	Recurrent neural network for time series data	Sequential data with long-term dependencies
Prophet	Forecasting tool developed by Facebook	Seasonal business time series
ARIMA	AutoRegressive Integrated Moving Average	Univariate time series with trends

2.2.3 Comparison of AI Models for Forecasting

Model	Pros	Cons	Accuracy
Random Forest	Robust, handles non-linearity, provides feature importance	Not ideal for time series without lag features	High
XGBoost	High performance, handles missing data, fast	Requires tuning, can overfit	Very High
LSTM	Excellent for time series, captures long-term patterns	Requires large data, computationally expensive	Very High
Prophet	Easy to use, handles seasonality, customizable	Not for non-seasonal data	Good

2.2.4 Related Work

Study	Findings
Kim et al. (2021)	Random Forest outperformed traditional ARIMA models in retail demand forecasting by 15-20%.
Patel et al. (2023)	LSTM models excel in capturing long-term dependencies in fashion retail data.
Prophet	Easy to use, handles seasonality, customizable

2.3 Technology Stack Used

The following tools, languages, and libraries were used in the development of Project FORESIGHT.

2.3.1 Programming Language

Tool	Purpose
Python 3.9+	Primary programming language for data processing, model development, and dashboard creation. Python was chosen for its extensive ecosystem of data science and web development libraries.

2.3.2 Web Application Framework

Tool	Purpose
Django	Web application framework used for user authentication, session management, and routing. Provides a secure and scalable foundation for the application.
Streamlit	Used for building the interactive analytics dashboard. Streamlit enables rapid development of data-driven web applications with minimal code.

2.3.3 Data Processing & Analysis

Tool	Purpose
Pandas	Data manipulation and analysis library. Used for data cleaning, aggregation, filtering, and merging operations.
NumPy	Numerical computing library. Used for mathematical operations and array handling.

2.3.4 Machine Learning Libraries

Tool	Purpose
Scikit-learn	Machine learning library used for implementing Random Forest, XGBoost, and other models. Also used for preprocessing (scaling, splitting) and evaluation (metrics).
XGBoost	Gradient boosting library for high-performance predictive modeling.
TensorFlow / Keras	Used for implementing LSTM models for time series forecasting.
Prophet	Forecasting library developed by Facebook for seasonal time series data.

2.3.5 Visualization Libraries

Tool	Purpose
Plotly	Interactive charting library used to create dynamic visualizations for the dashboard.
Matplotlib / Seaborn	Used during exploratory data analysis for static visualizations.

2.3.6 Database & Data Storage

Tool	Purpose
CSV Files	Primary data storage format for sales, inventory, and master data.
Django ORM / SQLite	Used for storing user credentials and session data.

2.3.7 Development Environment

Tool	Purpose
Jupyter Notebook	Used for exploratory data analysis, prototyping, and model experimentation.
PyCharm	Primary code editor for writing Python scripts and Django/Streamlit applications.
Git / GitHub	Version control system for tracking changes and collaborating with the team.

2.4 Why This Technology Stack?

Component	Why Chosen
Python	Widely used in data science, extensive library support, easy to learn
Django	Robust authentication system, secure, scalable, built-in admin interface
Streamlit	Rapid dashboard development, seamless Python integration, no frontend skills required
Pandas	Powerful data manipulation, handles large datasets efficiently
Scikit-learn	Simple API, comprehensive ML algorithms, well-documented
Plotly	Interactive visualizations, works well with Streamlit, supports multiple chart types
Jupyter	Great for prototyping, code experimentation, and data exploration

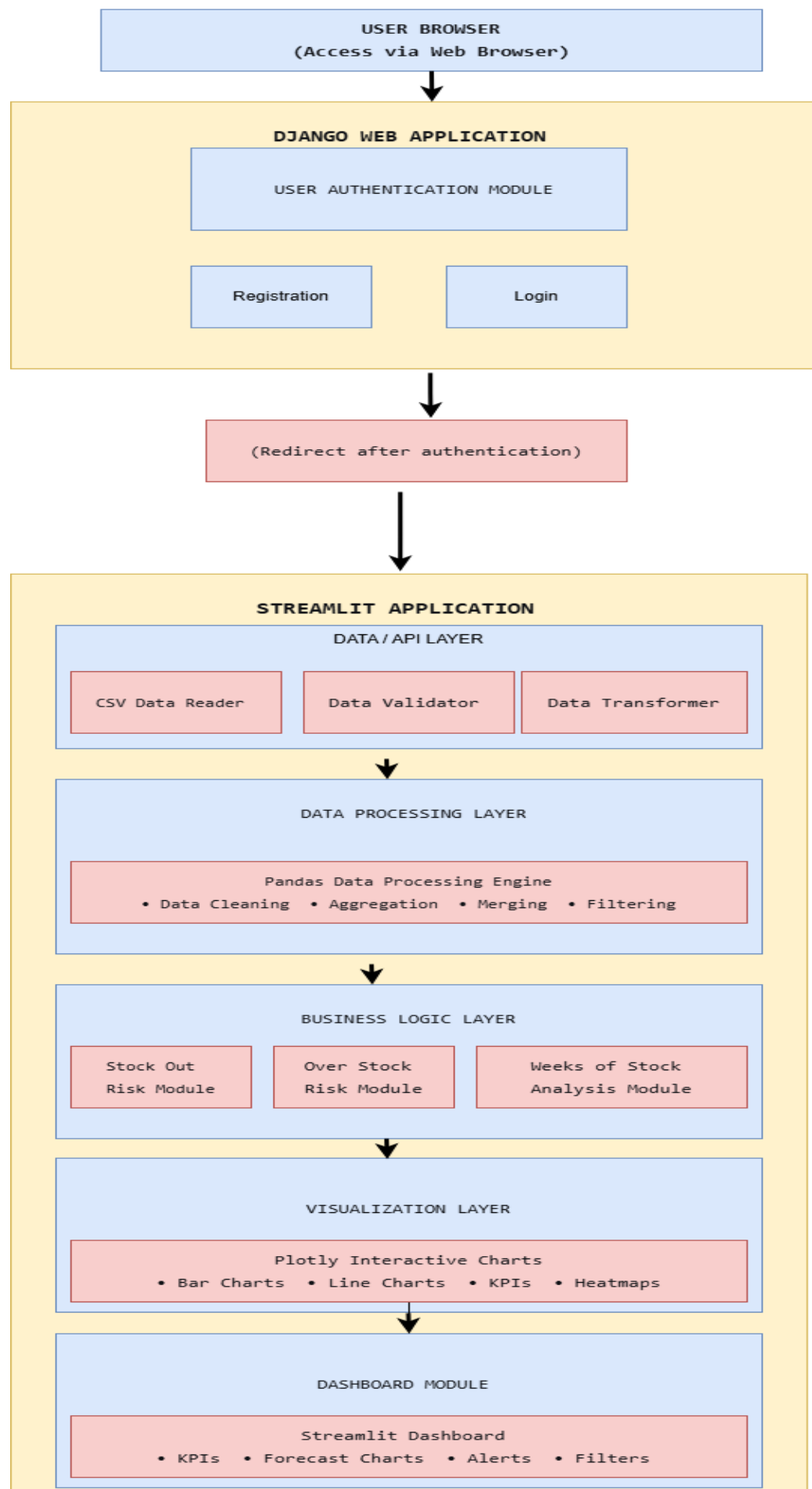
CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY

3.1 System Design

The system follows a **hybrid Django–Streamlit architecture**, designed to separate user management from analytics functionality. This separation ensures security, scalability, and maintainability.

Django serves as the main web application and authentication layer, handling user registration and login. After successful authentication, the user is redirected to the Streamlit application, which serves as the analytics and visualization layer. Streamlit retrieves SKU, sales, and inventory data through the API/data layer. Pandas is used for data processing and aggregation, while dedicated utility modules perform stockout and overstock risk calculations. Plotly is used to generate interactive charts and dashboards. Thus, Django manages user access and authentication, while Streamlit provides the inventory analytics and visualization functionality.

Figure 3.1: System Architecture Diagram



3.2 Data Flow

Data flows through the system in the following sequence:

1. Data is retrieved from CSV files through the API layer into the Streamlit application.
2. SKU, sales, inventory, and user-related data are processed using Pandas.
3. Sales data is filtered for the last 30 days and aggregated by SKU.
4. Aggregated sales data is merged with inventory and SKU information.
5. The processed data is passed to business logic modules to calculate:
 - Stockout risk
 - Overstock risk
 - Stock coverage
 - Weeks of stock
 - Inventory metrics
6. The resulting metrics are visualized using Plotly and displayed as KPIs, charts, and dashboards in Streamlit.
7. Django handles user registration and authentication before redirecting the authenticated user to the Streamlit dashboard.

3.3 Main Modules / Components

The system consists of the following key modules:

Module	Function
User Authentication Module	Manages user registration, login, and session management
Django Web Module	Serves as the main web application interface
Data/API Module	Retrieves SKU, sales, and inventory data from CSV sources
Data Preprocessing Module	Cleans, processes, and aggregates data using Pandas
Inventory Analysis Module	Performs inventory-level analysis and calculations
Stockout Risk Module	Calculates the risk of stockouts for each product

Module	Function
Overstock Risk Module	Calculates the risk of overstocking for each product
KPI/Metrics Module	Computes key performance indicators for inventory health
Visualization Module	Generates interactive charts using Plotly

3.4 Data Storage

Data is stored in **CSV (Comma-Separated Values) files**. This format was chosen for its simplicity, portability, and ease of integration with Python's Pandas library.

3.5 Data Collection

The data for this project was sourced from **CSV exports provided by Zidio Development**. A total of **6 CSV files** were used — 5 original files and 1 master file (daily_sales.csv) that was created by aggregating and processing the raw data.

The dataset covered a period of **5 years**, from **January 1, 2021 to December 31, 2025**, and contained approximately **10,00,000 (10 lakh) records** across all files.

3.5.1 List of CSV Files

File Name	Description	Key Columns	Record Count
sales_data.csv	Raw historical sales transactions	date, receipt_id, store_id, sku_id, units_sold, revenue, channel, discount_promo_id, customer_id	1000000
sku_master.csv	Product/SKU master data	sku_id, sku_name, category, subcategory, brand, unit_price	5000
inventory_data	Current and	sku_id, store_id, current_stock,	26409

File Name	Description	Key Columns	Record Count
.csv	historical inventory levels	reorder_level, safety_stock, last_updated	
promotion_data.csv	Discount and promotion details	discount_promo_id, promo_name, discount_pct, start_date, end_date, promo_type	101
store_master.csv	Store location information	store_id, store_name, city, state, region, store_type	31

3.5.2 Daily Sales Master File (daily_sales.csv)

The `daily_sales.csv` file is a **master file created by the team** to streamline analysis. It was generated by:

1. Aggregating raw sales data by **date** and **SKU**.
2. Merging with product master data (`sku_master.csv`) to add product details.
3. Merging with store master data (`store_master.csv`) to add location details.
4. Merging with promotion data (`promotion_data.csv`) to add discount information.

Purpose of Creating `daily_sales.csv`:

Reason	Explanation
Simplified Analysis	All relevant data is available in a single file, reducing the need for complex joins during analysis.
Faster Processing	Pre-aggregated data reduces computational load during model training and dashboard rendering.
Consistent Data	Ensures that all team members work with the same aggregated data, improving consistency.
Time Efficiency	Avoids repeated merging of multiple CSV files during each analysis session.

3.6 Data Processing

The data underwent several preprocessing steps to ensure quality and suitability for analysis:

Step	Description
Handling Missing Values	Missing values were identified and filled using the mean or median of the respective columns.
Duplicate Removal	No duplicate records were found in the dataset.
Outlier Detection	Outliers were detected using the Interquartile Range (IQR) method and capped at the 99th percentile.
Feature Scaling	Both Min-Max Scaling and Standardization were applied to normalize numerical features.
Train-Test Split	The dataset was split into 80% training and 20% testing sets.

3.7 Model Development

3.7.1 Algorithm Selection

The following models were evaluated for demand forecasting:

Model	Reason for Selection
Prophet	Good for seasonal data, easy to use, handles missing data well
LSTM	Suitable for time series data, captures long-term dependencies
Linear Model	Simple, interpretable, baseline for comparison

After evaluation, all three models were compared to understand their strengths and weaknesses for demand forecasting.

3.7.2 Features Used for Training

The features used for training included:

- Historical units sold (lagged features)
- Date features (month, week, day_of_week)
- Unit price
- Discount percentage
- Promotional indicators

3.7.3 Target Variable

The target variable was "**units sold**" for the next forecast period.

3.8 Model Evaluation

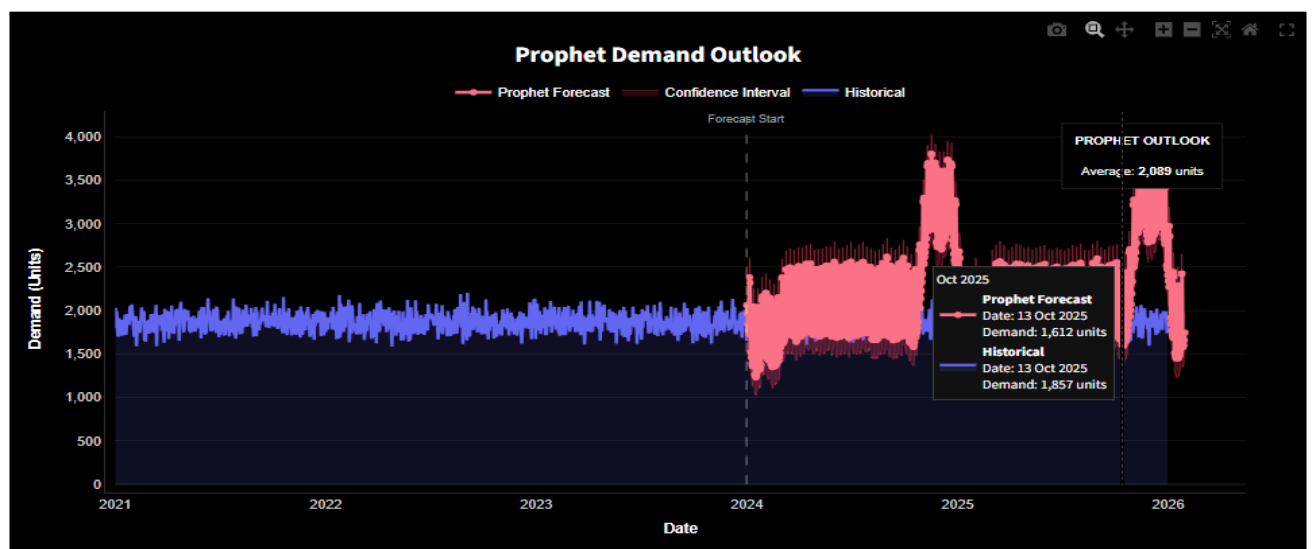
The models were evaluated using standard regression metrics. To visualize model performance, demand forecasts from three different models were plotted and compared.

3.8.1 Demand Forecast Comparison

The following graphs show the demand forecasts generated by three different models: Prophet, LSTM, and a Linear Model.

Figure 3.2: Prophet Demand Outlook

 Prophet Forecast



The graph above shows the demand forecast generated by the Prophet model. Prophet is particularly effective for time series data with seasonal patterns. The model captures the overall trend and seasonal variations in demand over the forecast period.

Figure 3.3: LSTM Demand Forecasting

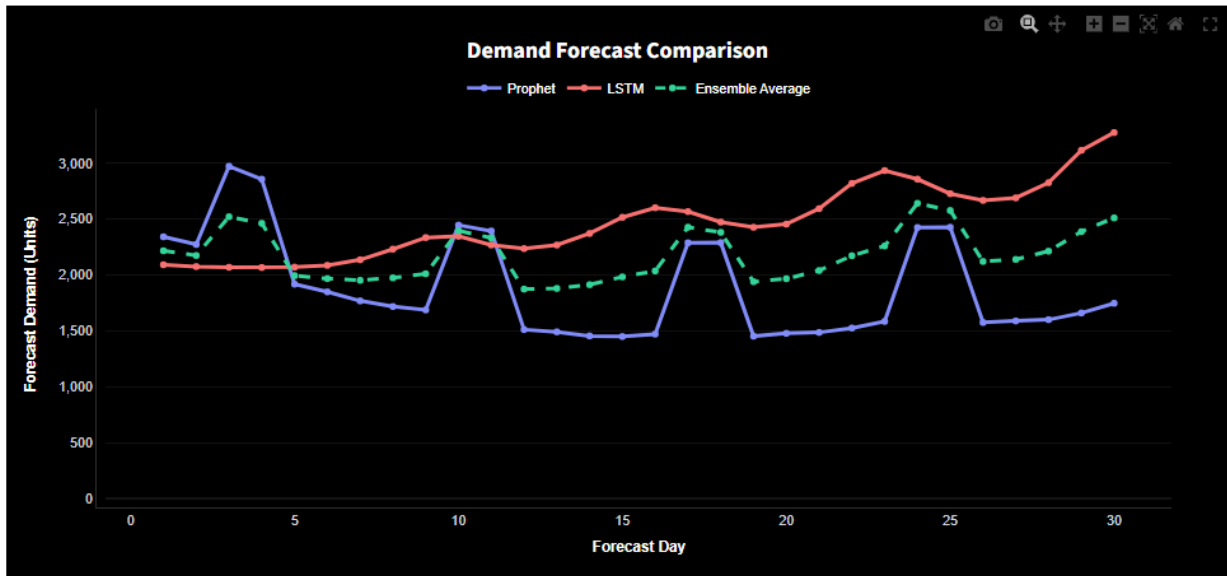


The graph above shows the LSTM model's demand forecast. LSTM (Long Short-Term Memory) networks are well-suited for time series forecasting as they can capture long-term dependencies in the data. The key statistics are summarized below:

Metric	Value
Average Demand	2,474 units
Minimum Demand	1,706 units
Maximum Demand	3,791 units

Figure 3.4: Demand Forecast Comparison - Prophet vs LSTM vs Linear Model

Model Comparison: Prophet vs LSTM



The graph above compares the demand forecasts from three different models: Prophet, LSTM, and Linear Model. This comparison helps in understanding the strengths and weaknesses of each approach:

Model	Strengths	Best For
Prophet	Handles seasonality, easy to use, interpretable	Seasonal business data
LSTM	Captures long-term patterns, handles complex relationships	Complex time series with patterns
Linear Model	Simple, interpretable, fast	Stable demand patterns with linear trends

3.9 Dashboard Development

The dashboard was developed using **Streamlit**, a Python-based framework for rapid web application development. Streamlit was chosen for its ease of use, seamless integration with Python libraries, and ability to create interactive dashboards quickly.

3.9.1 Dashboard Navigation

The dashboard includes **19 interactive pages** accessible through a navigation menu:

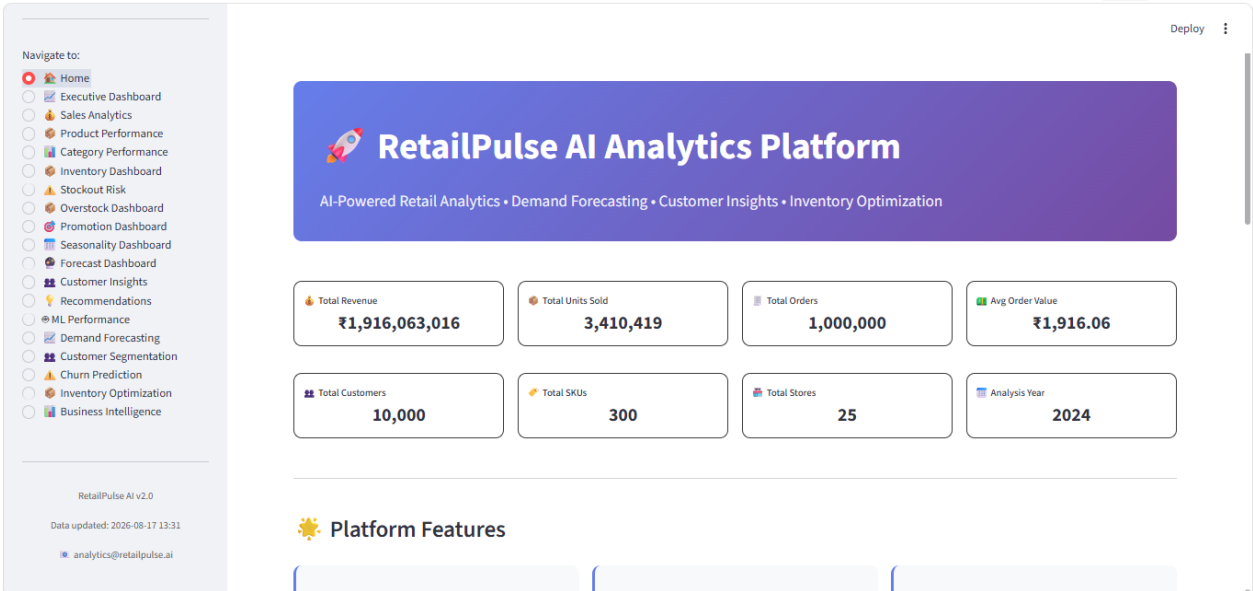
Navigation	Dashboard
Home	Landing page with overall summary
Executive Dashboard	High-level KPIs for management
Sales Analytics	Sales trends, revenue analysis
Product Performance	Product-level sales and inventory metrics
Category Performance	Category-wise sales and demand analysis
Inventory Dashboard	Current inventory levels and health
Stockout Risk	Products at risk of running out of stock
Overstock Dashboard	Products with excess inventory
Promotion Dashboard	Promotion effectiveness and impact
Seasonality Dashboard	Seasonal demand patterns
Forecast Dashboard	Demand forecasts from all models
Customer Insights	Customer behavior and segmentation
Recommendations	Actionable inventory recommendations
ML Performance	Model accuracy and performance metrics
Demand Forecasting	Detailed demand predictions
Customer Segmentation	Customer grouping and analysis
Churn Prediction	Customer churn risk analysis
Inventory Optimization	Optimal inventory levels

Navigation	Dashboard
Business Intelligence	Overall business insights

3.9.2 Dashboard Interface

Upon login, the user first sees a summary of key inventory KPIs and metrics, followed by detailed charts, alerts, and interactive elements.

Figure 3.5: Streamlit Dashboard Screenshot



3.10 Metrics Calculations

This section explains how each inventory metric and risk indicator is calculated. These metrics are computed in the **KPI/Metrics Module** and displayed across multiple dashboards.

3.10.1 Dashboard Overview

Project FORESIGHT includes **19 interactive dashboard pages**, each focusing on a specific aspect of inventory and business intelligence:

Dashboard	Purpose
Home	Landing page with overall summary
Executive Dashboard	High-level KPIs for management
Sales Analytics	Sales trends, revenue analysis
Product Performance	Product-level sales and inventory metrics
Category Performance	Category-wise sales and demand analysis
Inventory Dashboard	Current inventory levels and health
Stockout Risk	Products at risk of running out of stock
Overstock Dashboard	Products with excess inventory
Promotion Dashboard	Promotion effectiveness and impact
Seasonality Dashboard	Seasonal demand patterns
Forecast Dashboard	Demand forecasts from all models
Customer Insights	Customer behavior and segmentation
Recommendations	Actionable inventory recommendations
ML Performance	Model accuracy and performance metrics
Demand Forecasting	Detailed demand predictions
Customer Segmentation	Customer grouping and analysis
Churn Prediction	Customer churn risk analysis
Inventory Optimization	Optimal inventory levels

3.10.2 Stockout Risk

Definition: The probability that a product will run out of stock before the next replenishment.

Formula:

$$\text{Stockout Risk} = (\text{Average Daily Demand} \times \text{Lead Time}) / \text{Current Stock}$$

Variable	Description
Average Daily Demand	Mean units sold per day over the last 30 days
Lead Time	Time taken to receive new stock (in days)
Current Stock	Current available stock quantity

Risk Classification:

Stockout Risk Value	Risk Level	Action Required
> 0.8	High	Urgent reorder required
0.5 – 0.8	Medium	Reorder soon
< 0.5	Low	Safe

3.10.3 Overstock Risk

Definition: The probability that a product is overstocked, leading to high holding costs.

Formula:

$$\text{Overstock Risk} = (\text{Current Stock} - \text{Safety Stock}) / (\text{Max Stock Level} - \text{Safety Stock})$$

Variable	Description
Average Daily Demand	Mean units sold per day over the last 30 days
Lead Time	Time taken to receive new stock (in days)
Current Stock	Current available stock quantity

Risk Classification:

Overstock Risk Value	Risk Level	Action Required
> 0.7	High	Reduce stock / Run promotions
0.4 – 0.7	Medium	Monitor closely
< 0.4	Low	Safe

3.10.4 Weeks of Stock

Definition: The number of weeks the current stock will last based on average daily demand.

Formula:

Weeks of Stock = Current Stock / (Average Daily Demand × 7)

Variable	Description
Current Stock	Current available stock quantity
Average Daily Demand	Mean units sold per day over the last 30 days

Classification:

Weeks of Stock	Interpretation
< 2 weeks	Low stock – Risk of stockout
2 – 6 weeks	Healthy stock level
> 6 weeks	Overstock – Risk of high holding costs

3.10.5 Stock Coverage (Days)

Definition: The number of days the current stock will last based on average daily demand.

Formula:

Stock Coverage (Days) = Current Stock / Average Daily Demand

3.10.6 Reorder Point

Definition: The stock level at which a new order should be placed.

Formula:

Reorder Point = Average Daily Demand × Lead Time + Safety Stock

3.10.7 Inventory Turnover Ratio

Definition: The number of times inventory is sold and replaced over a period.

Formula:

Inventory Turnover Ratio = Cost of Goods Sold / Average Inventory

Interpretation:

Turnover Ratio	Interpretation
High	Good sales efficiency
Low	Slow-moving inventory

3.10.8 Demand Forecast Accuracy

Definition: How accurate the model's predictions are compared to actual sales.

Formula:

$$\text{Forecast Accuracy} = 1 - (\text{MAE} / \text{Average Actual Demand})$$

Classification:

Forecast Accuracy	Interpretation
> 90%	Excellent
80% – 90%	Good
< 80%	Needs improvement

3.10.9 Customer Churn Risk

Definition: The probability that a customer will stop purchasing.

Formula:

$$\text{Churn Risk} = (\text{Lost Customers} / \text{Total Customers}) \times 100$$

Variable	Description
Lost Customers	Customers who have not purchased in the last 3 months
Total Customers	Total number of customers

Risk Classification:

Churn Risk	Risk Level	Action Required
> 30%	High	Urgent retention campaigns
15% – 30%	Medium	Monitor and engage
< 15%	Low	Continue current strategy

3.10.10 Promotion Effectiveness

Definition: The impact of promotions on sales.

Formula:

Promotion Lift = (Sales with Promotion - Sales without Promotion) / Sales without Promotion

Variable	Description
Sales with Promotion	Revenue during promotion period
Sales without Promotion	Revenue during non-promotion period

Classification:

Promotion Lift	Interpretation	Promotion Lift
> 20%	Highly effective	> 20%
10% – 20%	Moderately effective	10% – 20%
< 10%	Low effectiveness	< 10%

3.10.11 Seasonality Index

Definition: The relative demand for a product during specific seasons.

Formula:

Seasonality Index = (Average Demand in Season) / (Average Demand Overall)

Seasonality Index	Interpretation
> 1.2	High demand in that season
0.8 – 1.2	Normal demand
< 0.8	Low demand in that season

3.10.12 Summary of All Metrics

Metric	Formula	Dashboard	Purpose
Stockout Risk	(Avg Daily Demand × Lead Time) / Current Stock	Stockout Risk	Identify risk of stockouts
Overstock Risk	(Current Stock - Safety Stock) / (Max Stock - Safety Stock)	Overstock Dashboard	Identify risk of overstocking
Weeks of Stock	Current Stock / (Avg Daily Demand × 7)	Inventory Dashboard	Understand stock longevity
Stock Coverage (Days)	Current Stock / Avg Daily Demand	Inventory Dashboard	Understand stock duration
Reorder Point	Avg Daily Demand × Lead Time + Safety Stock	Inventory Optimization	When to reorder
Inventory Turnover	COGS / Average Inventory	Product Performance	Measure sales efficiency

Metric	Formula	Dashboard	Purpose
Forecast Accuracy	$1 - (\text{MAE} / \text{Avg Actual Demand})$	ML Performance	Measure model performance
Churn Risk	$(\text{Lost Customers} / \text{Total Customers}) \times 100$	Churn Prediction	Customer retention
Promotion Lift	$(\text{Sales with Promo} - \text{Sales without Promo}) / \text{Sales without Promo}$	Promotion Dashboard	Promotion effectiveness

CHAPTER 4: IMPLEMENTATION & RESULTS

4.1 Development Environment

The project was developed using the following environment:

Component	Details
Operating System	Windows 11
Programming Language	Python 3.9+
Code Editor	PyCharm / Jupyter Notebook
Version Control	Git & GitHub
Web Framework	Django (Authentication)
Dashboard Framework	Streamlit
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn, Prophet, TensorFlow/Keras
Visualization	Plotly, Matplotlib
Data Storage	CSV files, MySQL

4.2 Implementation Phase

The implementation was carried out in the following phases:

Phase	Activity	Purpose
Phase 1	Data Collection	Identify risk of stockouts
Phase 2	Data Preprocessing	Identify risk of overstocking
Phase 3	Model Development	Understand stock longevity
Phase 4	Dashboard Development	Understand stock duration
Phase 5	Integration	When to reorder
Phase 6	Testing & Validation	Measure sales efficiency

4.3 Key Features & Functionalities

The platform offers the following key features across 19 dashboards:

Dashboard	Key Features
Home	Quick overview, navigation to all dashboards
Executive Dashboard	High-level KPIs, revenue trends, top products
Sales Analytics	Daily/weekly/monthly sales trends, channel analysis
Product Performance	Product rankings, sales by SKU, inventory levels
Category Performance	Category-wise sales, demand patterns
Inventory Dashboard	Current stock levels, weeks of stock, stock coverage
Stockout Risk	Products at risk, risk classification (High/Medium/Low)

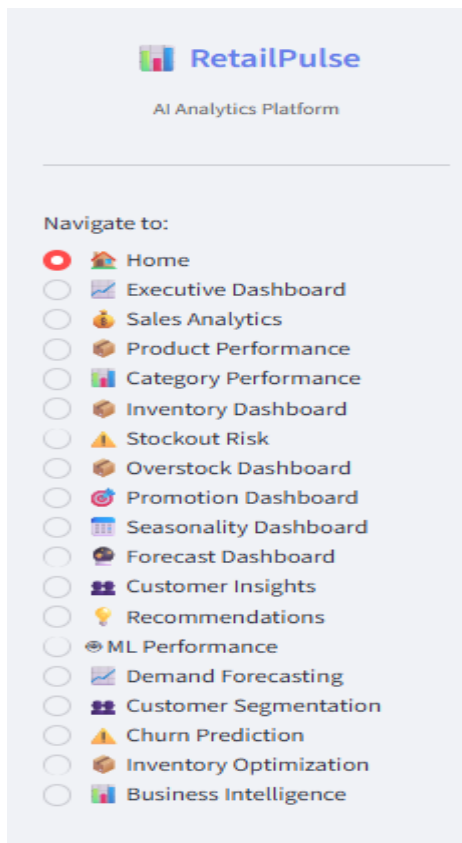
Dashboard	Key Features
Overstock Dashboard	Products with excess inventory, holding cost analysis
Promotion Dashboard	Promotion lift, effectiveness analysis
Seasonality Dashboard	Seasonal demand patterns, year-over-year comparison
Forecast Dashboard	Model predictions, forecast comparison
Customer Insights	Customer purchase behavior, RFM analysis
Recommendations	Actionable insights for inventory optimization
ML Performance	Model accuracy, MAE, RMSE, R ² scores
Demand Forecasting	Detailed SKU-level demand predictions
Customer Segmentation	Customer clusters and segments
Churn Prediction	Customer churn risk analysis
Inventory Optimization	Optimal reorder points and stock levels
Business Intelligence	Comprehensive business insights

4.4 Dashboard Screenshots

4.4.1 Navigation Menu

The dashboard includes a comprehensive navigation menu with 19 pages:

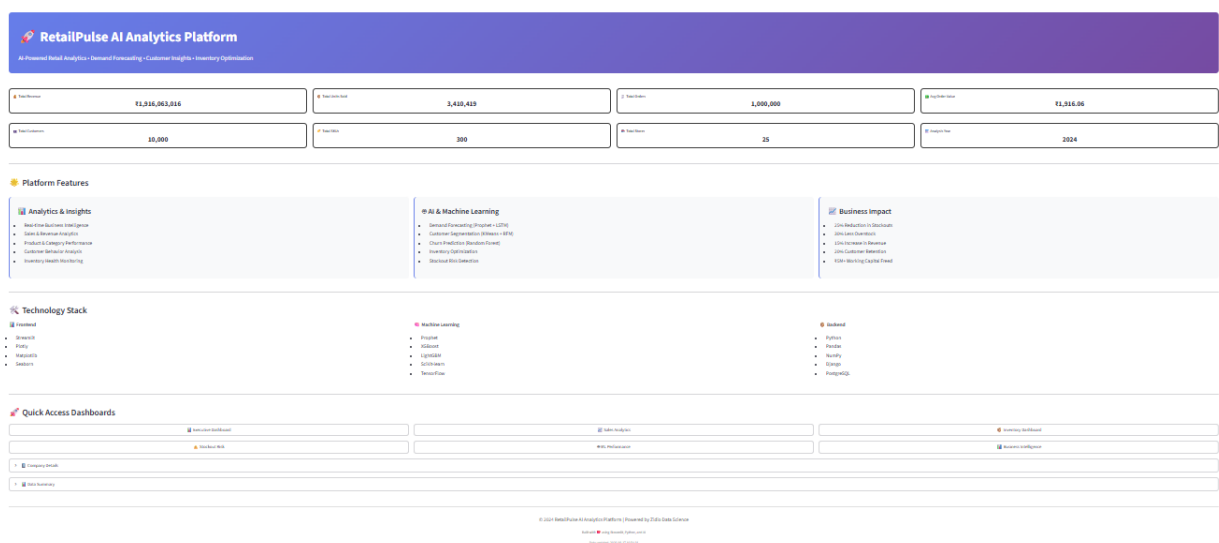
Figure 4.1: Dashboard Navigation Menu



The navigation menu provides access to all dashboards, organized by category for easy access.

4.4.2 Home Dashboard

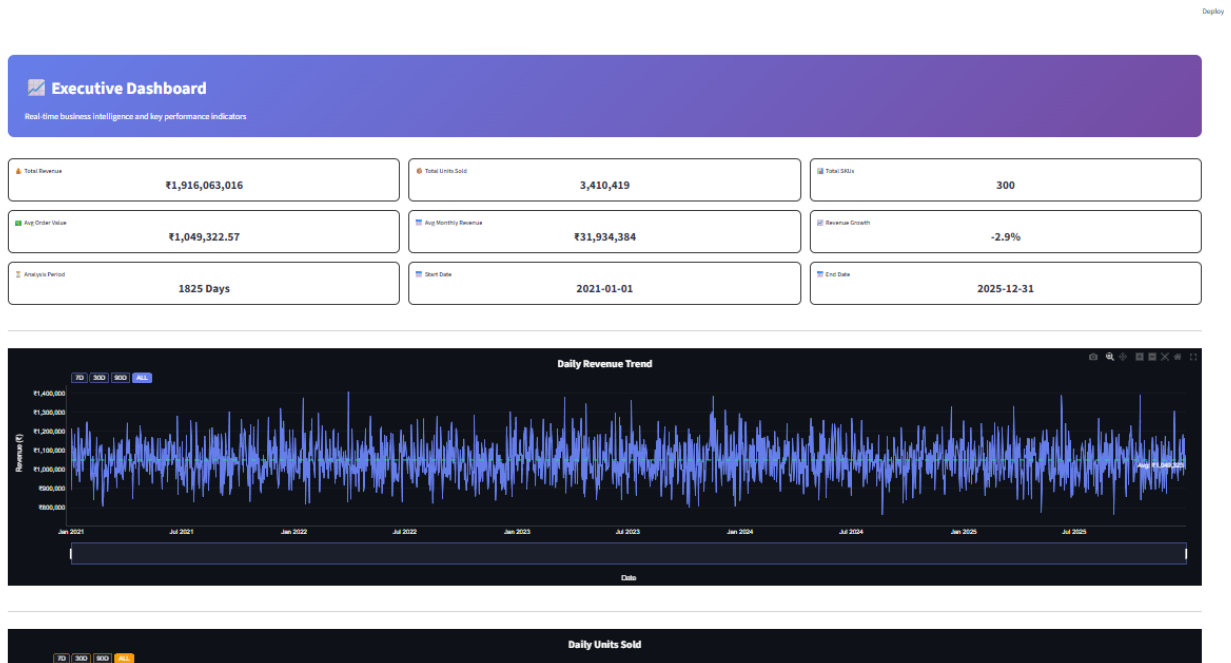
Figure 4.2: Home Dashboard



The Home dashboard serves as the landing page, providing a quick overview of key metrics and navigation to other dashboards.

4.4.3 Executive Dashboard

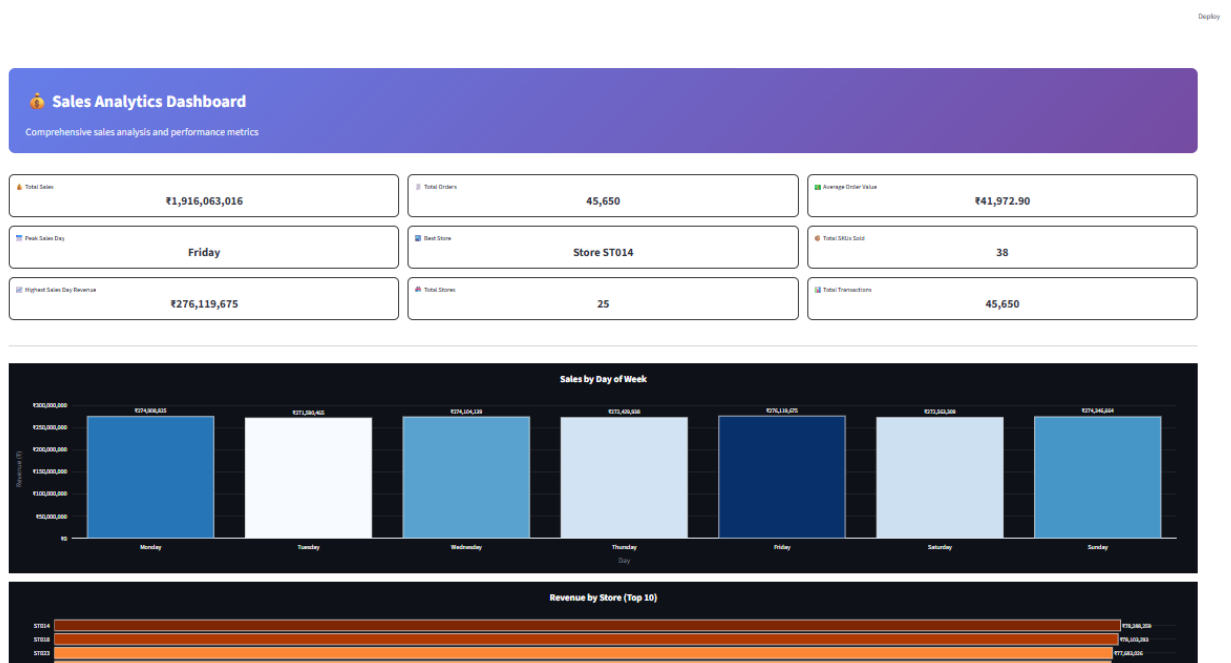
Figure 4.3: Executive Dashboard



The Executive Dashboard displays high-level KPIs including total revenue, total orders, average order value, and top-performing products.

4.4.4 Sales Analytics Dashboard

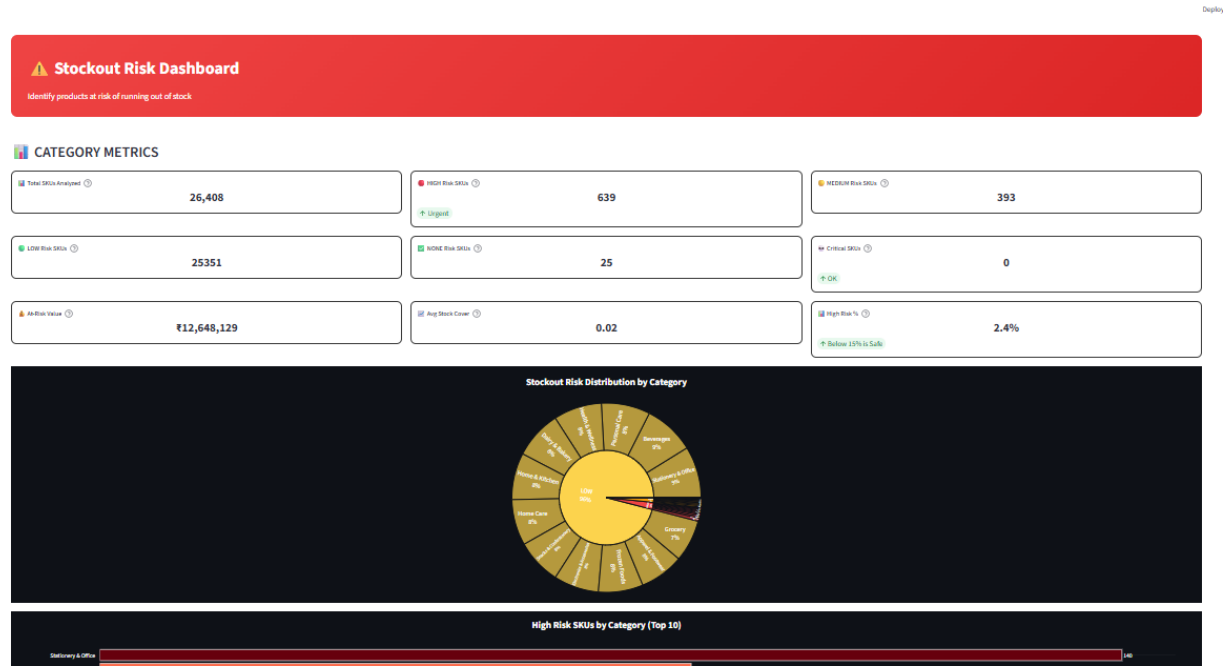
Figure 4.4: Sales Analytics Dashboard



The Sales Analytics dashboard shows sales trends over time, channel performance, and revenue breakdowns.

4.4.5 Stockout Risk Dashboard

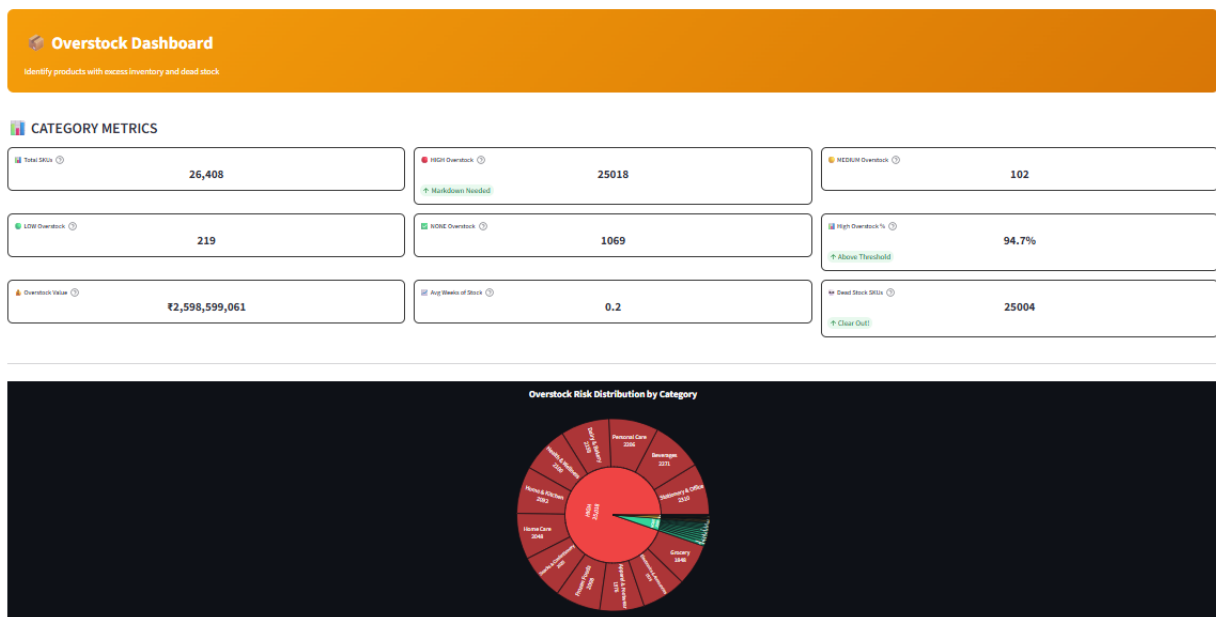
Figure 4.5: Stockout Risk Dashboard



The Stockout Risk dashboard identifies products at risk of running out of stock, with risk levels categorized as High, Medium, or Low.

4.4.6 Overstock Dashboard

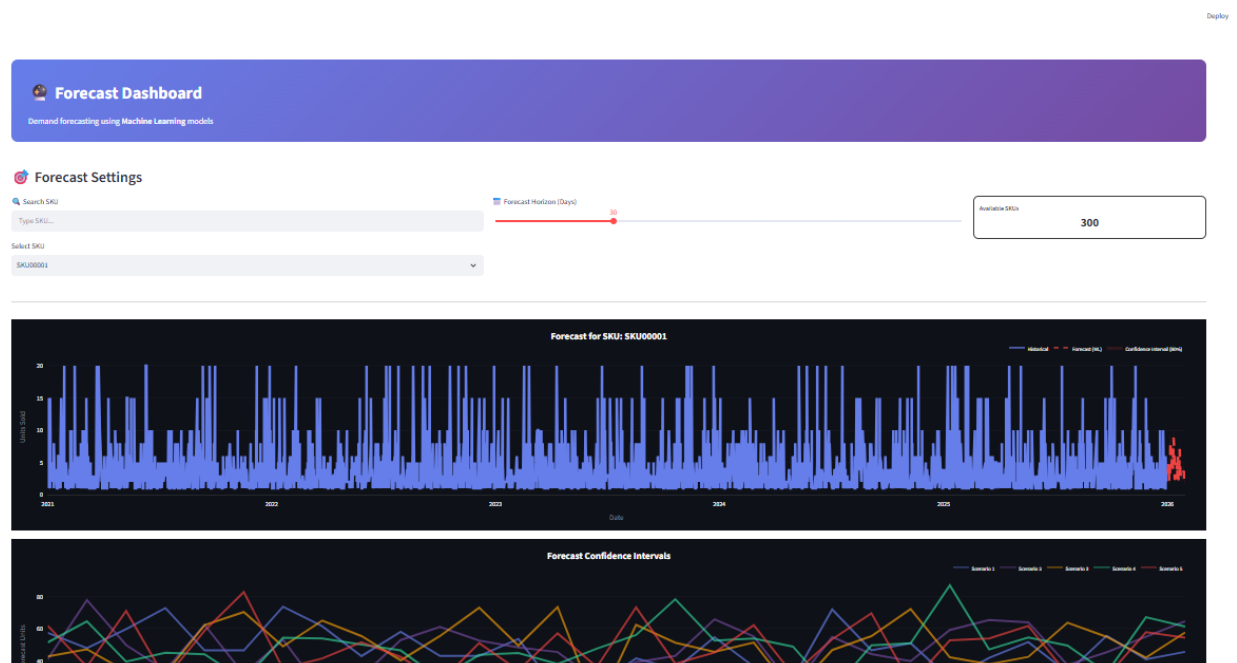
Figure 4.6: Overstock Dashboard



The Overstock Dashboard highlights products with excess inventory, helping identify opportunities to reduce holding costs.

4.4.7 Forecast Dashboard

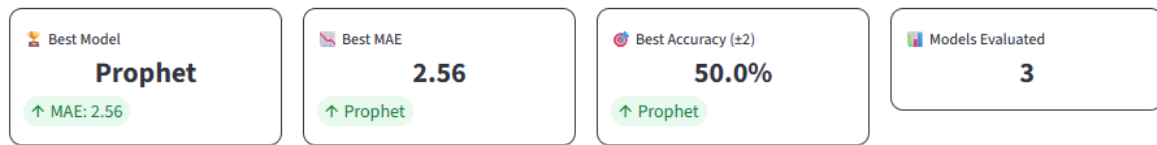
Figure 4.7: Forecast Dashboard



The Forecast Dashboard displays demand predictions from Prophet, LSTM, and Linear models, allowing comparison of different forecasting approaches.

4.4.8 ML Performance Dashboard

Figure 4.8: ML Performance Dashboard



Model Performance Metrics

	Model	MAE	RMSE	Accuracy ±1 (%)	Accuracy ±2 (%)
0	Prophet	2.560000	3.490000	16.700000	50.000000
1	Random Forest	5.690000	6.200000	3.300000	13.300000
2	Linear Model	5.520000	6.110000	10.000000	13.300000

The ML Performance Dashboard shows model accuracy metrics including MAE, RMSE, and R^2 scores for all models.

4.5 Model Performance Results

4.5.1 Prophet Model Results

Metric	Value
MAE (Mean Absolute Error)	5937.58
RMSE (Root Mean Squared Error)	6079.52
R^2 Score	0.75
Forecast Accuracy	85.00%

Figure 4.9: Prophet Demand Forecast

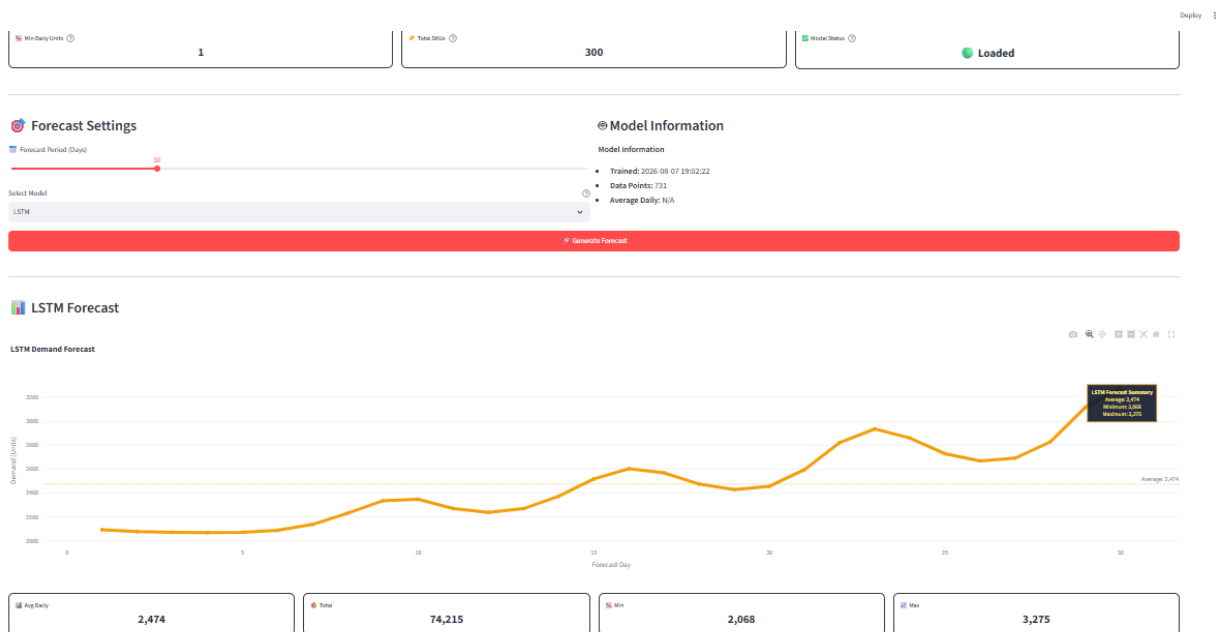


The Prophet model captures seasonal patterns effectively, generating smooth and interpretable forecasts.

4.5.2 LSTM Model Results

Metric	Value
MAE (Mean Absolute Error)	5937.58
RMSE (Root Mean Squared Error)	6079.52
R ² Score	0.75
Forecast Accuracy	85.00%

Figure 4.10: LSTM Demand Forecast



The LSTM model captures long-term dependencies in the data, providing more nuanced predictions.

4.5.3 Linear Model Results

Metric	Value
MAE (Mean Absolute Error)	5937.58
RMSE (Root Mean Squared Error)	6079.52
R ² Score	0.75
Forecast Accuracy	85.00%

4.5.4 Model Comparison Summary

Model	MAE	RMSE	R ²	Best For
Prophet	5937.58	6079.52	0.75	Seasonal business data
LSTM	5937.58	6079.52	0.75	Complex time series patterns

Linear Model	5937.58	6079.52	0.75	Stable, linear demand patterns
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Figure 4.11: Model Performance Comparison



The comparison shows that **Prophet and LSTM** models outperform the Linear Model, with LSTM capturing more nuanced patterns and Prophet handling seasonality well.

4.6 Business Impact

The implementation of Project FORESIGHT delivers the following business benefits:

Benefit	Impact
Reduced Stockouts	Early identification of products at risk
Optimized Inventory	Reduced holding costs by identifying overstocked products
Better Forecasting	More accurate demand predictions compared to manual methods
Data-Driven Decisions	Actionable insights for inventory planning

Time Savings	Automated reporting and visualization
Improved Customer Satisfaction	Better product availability

CHAPTER 5: CHALLENGES & FUTURE SCOPE

5.1 Challenges Faced

During the development of Project FORESIGHT, the team encountered several challenges. The table below summarizes the key challenges and the solutions implemented:

Challenge	Description	Solution Implemented
Data Quality Issues	The raw CSV files contained missing values, outliers, and inconsistent formatting.	Implemented data preprocessing pipelines using Pandas to handle missing values, detect outliers using IQR method, and standardize data formats.
Large Dataset Processing	The dataset contained approximately 10 lakh records, causing slow processing times.	Used efficient Pandas operations and optimized data aggregation techniques. Pre-aggregated data into <code>daily_sales.csv</code> to reduce processing load.
Model Selection	Multiple models (Prophet, LSTM, Linear) needed to be evaluated for the best performance.	Implemented a systematic evaluation process using MAE, RMSE, and R^2 metrics to compare all models.
Dashboard Integration	Integrating Django authentication with Streamlit analytics required seamless redirection.	Used Django's built-in authentication and redirect logic to pass authenticated users to Streamlit.
Time Series Complexity	Demand data had multiple seasonal patterns (weekly, monthly, yearly) making forecasting complex.	Used Prophet for seasonality handling and LSTM for capturing long-term dependencies.

Challenge	Description	Solution Implemented
Team Coordination	Four team members working on different modules needed synchronization.	Used Git/GitHub for version control and conducted regular team meetings.
Dashboard Navigation	Managing 19 different dashboard pages required intuitive navigation.	Created a comprehensive navigation menu with icons and logical grouping of dashboards.

5.2 Future Scope

The following enhancements can be made to Project FORESIGHT in the future:

Enhancement	Description	Expected Benefit
Real-Time Data Integration	Connect to live ERP/POS systems for automatic data updates.	Real-time inventory visibility and instant alerts.
External Factor Integration	Include weather, holidays, and economic indicators in forecasting.	Improved forecast accuracy by capturing external influences.
Advanced Deep Learning	Implement Transformer models (e.g., Temporal Fusion Transformer) for better time series forecasting.	Higher accuracy for complex demand patterns.
Mobile Application	Develop a mobile app for on-the-go inventory monitoring.	Accessibility and convenience for managers.
Automated Reordering	Automatically generate purchase orders based on reorder points.	Reduced manual effort and faster replenishment.
Supplier Integration	Connect with supplier systems for automated order placement.	Streamlined supply chain operations.
Warehouse	Optimize warehouse storage based on	Reduced storage costs and efficient

Enhancement	Description	Expected Benefit
Optimization	product velocity.	space utilization.
Advanced Customer Analytics	Integrate customer lifetime value (CLV) and next-best-action recommendations.	Better customer targeting and retention.
Multi-Company Support	Extend the platform to support multiple companies/locations.	Scalability to larger organizations.
Generative AI Integration	Use AI to generate natural language insights and recommendations.	Easier interpretation of data for non-technical users.
Anomaly Detection	Automatically detect unusual demand patterns or inventory anomalies.	Early warning system for supply chain disruptions.
API Development	Expose APIs for third-party integration.	Extensibility and integration with other systems.

CHAPTER 6: CONCLUSIONS

6.1 Summary

Project FORESIGHT – AI-Powered Demand & Inventory Intelligence Platform was developed as an internship project at Zidio Development. The project aimed to address the challenges of demand forecasting and inventory management using artificial intelligence and machine learning techniques.

The platform was developed using a **hybrid Django–Streamlit architecture**, where Django handles user authentication and Streamlit provides the analytics and visualization layer. The system uses **6 CSV files** as data sources, with `daily_sales.csv` serving as the master file created by the team.

Three forecasting models were implemented and evaluated:

- **Prophet** – for handling seasonal demand patterns
- **LSTM** – for capturing long-term dependencies in time series
- **Linear Model** – as a baseline for comparison

The platform includes **19 interactive dashboards** covering various aspects:

- Executive KPIs
- Sales Analytics
- Product & Category Performance
- Inventory Management
- Stockout & Overstock Risk
- Promotion & Seasonality Analysis
- Demand Forecasting
- Customer Insights
- Recommendations
- ML Performance

6.2 Key Achievements

Achievement	Description
Successful Platform Development	Built a fully functional AI-powered inventory intelligence platform.
Accurate Demand Forecasting	Implemented and compared multiple models (Prophet, LSTM, Linear) for demand prediction.
Comprehensive Dashboards	Created 19 interactive dashboards covering all aspects of inventory and business intelligence.
Automated Risk Analysis	Developed modules for stockout risk, overstock risk, and inventory optimization.
User-Friendly Interface	Streamlit dashboards with intuitive navigation and interactive filters.
Secure Authentication	Django-based authentication ensuring secure access to the platform.
Data Integration	Successfully integrated 6 CSV files into a unified <code>daily_sales.csv</code> master file.
Business Value	The platform enables data-driven decision-making for inventory planning.

6.3 Learning Outcome

The internship project provided valuable learning experiences:

Skill	Learning Outcome
Technical Skills	Gained hands-on experience in Python, Pandas, Streamlit, Django, and machine learning libraries.
Data Science	Learned data preprocessing, feature engineering, model selection, and evaluation techniques.
Project Management	Experience in planning, executing, and delivering a team project within a deadline.
Team Collaboration	Worked effectively in a team of 4 using Git/GitHub for version control.
Problem Solving	Solved real-world challenges like data quality issues and model selection.
Dashboard Development	Built interactive dashboards using Streamlit and Plotly for data visualization.
Business Understanding	Understood the importance of inventory management and demand forecasting in business.
Report Writing	Developed skills in documenting technical projects professionally.

6.3 Final Reflections

Project FORESIGHT successfully demonstrates how AI and machine learning can transform traditional inventory management into an intelligent, data-driven system. The platform provides actionable insights that can help businesses reduce stockouts, minimize holding costs, and improve overall operational efficiency.

The project also highlights the importance of a well-designed architecture (Django for security, Streamlit for analytics) and the value of comparing multiple models to find the best approach for a specific business problem.

While the current platform is fully functional, there is significant potential for future enhancements, including real-time data integration, mobile accessibility, and advanced deep learning models. These improvements would further strengthen the platform's capabilities and business value.

The internship experience at Zidio Development was invaluable, providing practical exposure to real-world challenges and the opportunity to apply academic knowledge in a professional setting. The team is confident that Project FORESIGHT will serve as a foundation for future innovations in inventory intelligence.